

Homework for Chap 10

Suppose the external electric field applied to a piece of metal is $E_x^0 e^{-i\omega t}$, using the Drude model to derive the complex conductivity $\sigma(\omega) = \sigma_0 \frac{1}{1-i\omega\tau}$, and get the expression for σ_0 .

$$F = -eE(t) = -e \cdot E_0 e^{-i\omega t}$$

Drude:

$$m \frac{dv}{dt} = -eE - \frac{mv}{\tau}$$

$$m \ddot{x} = F_x = -\frac{m}{\tau} \dot{x} + eE_0 e^{-i\omega t}$$

$$\ddot{x} + \frac{1}{\tau} \dot{x} = \frac{eE_0}{m} e^{-i\omega t}$$

$$x(t) = x_0 e^{-i\omega t}$$

$$(-\omega^2 - i\frac{\omega}{\tau}) x_0 = -\frac{e}{m} E_0$$

$$x_0 = \frac{eE_0}{m(\omega^2 + i\frac{\omega}{\tau})}$$

$$J = -en \dot{x}(t) = -en \frac{dv}{dt} = i\omega ne \frac{eE_0}{m(\omega^2 + i\frac{\omega}{\tau})} = \frac{ne^2 E_0 \tau}{m(1 - i\omega\tau)}$$

$$\sigma = \frac{J}{E} = \frac{ne^2 \tau}{m(1 - i\omega\tau)} = \sigma_0 \frac{1}{1 - i\omega\tau}$$